

Total no. of Pages:2

Roll no:

Degree: B. Tech. Semester: 3rd — Course work
END-SEMESTER EXAMINATION, Nov-Dec 2023

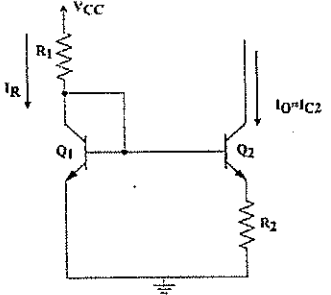
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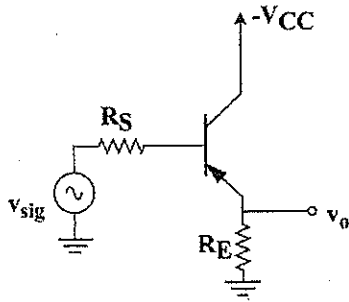
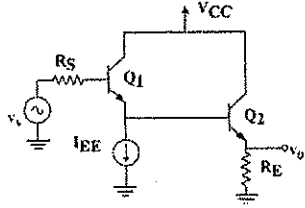
Duration: 3 Hours

Course Title: Microelectronics

Course Code: ECECC07

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q.no	Question	Marks	CO
Q1	Attempt any two parts of the following		
1(a)	For the Widlar current source shown in Fig. 1 derive the expression for output current I_o in terms of reference current I_R and other relevant parameters. Also, determine the values of R_1 and R_2 to give $I_o=5\mu A$ and $I_R=1mA$, given that $V_{cc}=30V$, $V_T=26mV$, $V_{BE}=0.7$ and $\beta=100$.	4	CO 1
	 Fig.1		
1(b)	A common collector stage shown in Fig. 2 is biased at $I_{CQ}=0.25mA$, $\beta=150$ and is driven by $3k\Omega$ source resistance R_S . (i) What value of R_E is needed to make $R_{out}=110\Omega$? (ii) Using the value of R_E obtained in (i) determine the overall voltage-gain V_o/V_{sig} and the input resistance R_{in} .	4	CO 1

	 Fig.2		
1(c)	For a common-source amplifier having an active load, determine the expressions for its input resistance R_{in} , output resistance R_{out} , and the voltage-gain A_v .	4	CO 1
Q2	Attempt any two parts of the following		
2(a)	Draw the circuit of a Cascode amplifier and explain its advantages. Also, draw its small signal equivalent circuit and therefrom, determine the expression for its output impedance.	4	CO 2
2(b)	The Darlington pair shown in Fig.3 is used as an emitter-follower with $R_E=500\Omega$ and is driven from a $50k\Omega$ source. The transistor used in the circuit has parameters as $\beta_o=150$ and $V_A=100$. The transistor Q_2 is biased at $1mA$ and Q_1 is biased at $15\mu A$. Find A_v , A_i , R_o and R_i .	4	CO 2
	 Fig.3		
2(c)	The differential amplifier shown in figure below uses transistors with $\beta=100$. Evaluate the following: (i) The input differential resistance R_{id} . (ii) The overall differential voltage gain V_{od}/V_{sig} (neglecting the effect of r_o).	4	CO 2

Q3	Attempt any two parts of the following		
3(a)	Draw the configurations for all the four basic feedback topologies and derive the expressions of R_{if} and R_{of} for the <i>shunt-shunt</i> feedback topology.	4	CO 3
3(b)	Draw and explain the circuit of the RC phase shift oscillator and derive the expression for the frequency of oscillation and condition to obtain sustained oscillations.	4	CO 3
3(c)	In the circuit shown below, determine the small-signal voltage gain V_o/V_s , the input resistance R_{in} , and the output resistance $R_{out} = R_{of}$. Assume that the transistor has $\beta=100$.	4	CO 3
Q4	Attempt any two parts of the following		

4(a)	Explain the architecture of a two-stage bipolar op-amp. Define and explain the significance of (i) Slew rate and (ii) PSRR, as applicable to op-amps.	4	CO 4																		
4(b)	Determine the input common-mode range and output swing of a two-stage CMOS op-amp. Also, explain the condition to eliminate systematic DC offset voltage in a two-stage CMOS op-amp.	4	CO 4																		
4(c)	In the circuit given below, $I_{REF} = 90\mu A$, $V_{tn} = 0.7$, $V_{tp} = -0.8$, $\mu_n C_{ox} = \frac{40\mu A}{V^2}$, V_A (for all devices) = $10V$, $V_{DD} = V_{SS} = 2.5V$ and aspect ratio of the MOS transistors are as given in Table I. <table border="1"> <caption>Table I</caption> <tr> <th>Transistor</th><th>Q₁</th><th>Q₂</th><th>Q₃</th><th>Q₄</th><th>Q₅</th><th>Q₆</th><th>Q₇</th><th>Q₈</th></tr> <tr> <th>W/L</th><td>20/0.8</td><td>20/0.8</td><td>5/0.8</td><td>5/0.8</td><td>40/0.8</td><td>10/0.8</td><td>40/0.8</td><td>40/0.8</td></tr> </table> Find the values of (i) the open-loop DC voltage gain and (ii) the output resistance.	Transistor	Q ₁	Q ₂	Q ₃	Q ₄	Q ₅	Q ₆	Q ₇	Q ₈	W/L	20/0.8	20/0.8	5/0.8	5/0.8	40/0.8	10/0.8	40/0.8	40/0.8	4	CO 4
Transistor	Q ₁	Q ₂	Q ₃	Q ₄	Q ₅	Q ₆	Q ₇	Q ₈													
W/L	20/0.8	20/0.8	5/0.8	5/0.8	40/0.8	10/0.8	40/0.8	40/0.8													
Q5	Attempt any two parts of the following																				
5(a)	Determine the maximum conversion efficiency of a Class A power amplifier.	4	CO 5																		
5(b)	For an ideal Class B push pull amplifier show that the collector dissipation P_c is zero at no signal ($V_m=0$) and rises as V_m increases and passes through a maximum $P_{c(max)}$ at $V_m = 2V_{CC}/\pi$.	4	CO5																		
5(c)	A class B push-pull amplifier is supplied with $V_{CC} = 50V$. The signal brings the collector voltage down to $V_{min}=5V$. The total dissipation from both transistors is $40W$. Find the total power and conversion efficiency.	4	CO5																		